GEN-AI HANDS-ON-1

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Observation:

The Transformers library serves as an interface connecting Python applications with AI models hosted on Hugging Face. It simplifies model usage through the pipeline() function, which automates key stages such as preprocessing, model inference, and postprocessing. We will be looking into two models –

1) distilGPT2

2) gpt2

seed (42)

Let's see what the model thinks Generative AI creates.

```
masked_sentence = "The goal of Generative AI is to create new [MASK]."
preds = mask_filler(masked_sentence)

for p in preds:
    print(f"{p['token_str']}: {p['score']:.2f}")
```

```
applications: 0.06
ideas: 0.05
problems: 0.05
systems: 0.04
information: 0.03
```

Step 3: Fast Model (`distilgpt2`)

Let's see how the smaller model performs.

```
# Initialize the pipeline with the specific model
fast_generator = pipeline('text-generation', model='distilgpt2')

# Generate text
output_fast = fast_generator(prompt, max_length=50, num_return_sequences=1)
print(output_fast[0]['generated_text'])
```

```
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for
Generative AI is a revolutionary technology that is designed to work with existing AI systems. It has been developed by the University of California, B


The research team led by Professor Daniel Kranz, from the University of California, Berkeley, has developed a program to learn how to use the AI to imp
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```

Step 4: Standard Model (gpt2)

Now let's try the standard model.

```
smart_generator = pipeline('text-generation', model='gpt2')

output_smart = smart_generator(prompt, max_length=50, num_return_sequences=1)
print(output_smart[0]['generated_text'])
```

```
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for
Generative AI is a revolutionary technology that allows users to build AI that can help solve complex problems. It brings together hundreds of differen

The AI is a model of human intelligence, and has many aspects that are similar to artificial intelligence. It can learn from humans, and it can adapt t

It is the main driving force behind the new Artificial Intelligence, and the AI is very important to the success of AI. The new AI is designed to work

The AI is designed to be scalable and adaptable to different environments. It can be used to solve complex problems without relying on humans. It can b

The new AI is designed to work out problems that need to be solved in a way that is
```

4.2 Question Answering

This task is **Extractive**. We provide a `context` (our text) and a `question`. The model highlights the answer within the text.

```
qa_pipeline = pipeline("question-answering", model="distilbert-base-cased-distilled-squad")
```

```
config.json: 100%  473/473 [00:00<00:00, 35.4kB/s]
model.safetensors: 100%  261M/261M [00:06<00:00, 27.3MB/s]
tokenizer_config.json: 100%  49.0/49.0 [00:00<00:00, 3.67kB/s]
vocab.txt: 100%  213k/213k [00:00<00:00, 6.29MB/s]
tokenizer.json: 100%  436k/436k [00:00<00:00, 6.56MB/s]
Device set to use cpu
```

Let's ask about the risks mentioned in our text.

```
questions = [
    "What is the fundamental innovation of the Transformer?",
    "What are the risks of using Generative AI?"
]

for q in questions:
    res = qa_pipeline(question=q, context=text[:5000])
    print(f"\nQ: {q}")
    print(f"A: {res['answer']}")
```

```
Q: What is the fundamental innovation of the Transformer?
A: to identify hidden patterns, structures, and relationships within the data

Q: What are the risks of using Generative AI?
A: data privacy, intellectual property, and academic integrity
```

seed(70)

Step 3: Fast Model (distilgpt2)

Let's see how the smaller model performs.

```
prompt = "Generative AI is a revolutionary technology that"
```

```
# Initialize the pipeline with the specific model
fast_generator = pipeline('text-generation', model='distilgpt2')

# Generate text
output_fast = fast_generator(prompt, max_length=50, num_return_sequences=1)
print(output_fast[0]['generated_text'])
```

```
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to max length. Defaulting to 'longest_first' truncat
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/tr
Generative AI is a revolutionary technology that advances the understanding of natural selection and the capacity for scientific research. As it turns out, the current AI technology is much more compl

In a study published in the Journal of Experimental Computational Biology (JETB), researchers from the University of Virginia, and researchers at University of Michigan, conducted in collaboration wit
```

distilGPT2 continues to provide fast responses; however, its outputs vary depending on the seed value used.

Step 4: Standard Model (gpt2)

Now let's try the standard model.

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output_smart = smart_generator(prompt, max_length=50, num_return_sequences=1)
print(output_smart[0]['generated_text'])
```

```
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to max length. Defaulting to 'longest_first' truncat
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Both `max_new_tokens` (=256) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/tr
Generative AI is a revolutionary technology that enables intelligent AI to replace human agents. The AI is based on the concept of natural selection. Natural selection by natural selection is an impor

The goal of these methods is to eliminate the natural selection bias in our society, which is the natural bias. The natural bias is a result of the cognitive, behavioral and psychological processes th

When we are presented with a choice, we recognize that we are the ones that are being selected. However, when we are presented with a choice, they are also the ones that are being selected. If the cho

In a free society, where we have choices about whether we are good or bad, that is, if we are good or bad, we are at risk for being selected.

What we should be doing is
```

While both outputs remain understandable, the wording, structure, and ordering of sentences generated with seed value 70 differ from those produced with seed value 42. The results generated using seed 42 appeared more coherent, with reduced repetition and improved logical flow.

4.2 Question Answering

This task is **Extractive**. We provide a `context` (our text) and a `question`. The model highlights the answer within the text.

```
qa_pipeline = pipeline("question-answering", model="distilbert-base-cased-distilled-squad")
```

```
Device set to use cpu
```

Let's ask about the risks mentioned in our text.

```
questions = [
    "What is the fundamental innovation of the Transformer?",
    "What are the risks of using Generative AI?"
]

for q in questions:
    res = qa_pipeline(question=q, context=text[:5000])
    print(f"\nQ: {q}")
    print(f"A: {res['answer']}")
```

```
Q: What is the fundamental innovation of the Transformer?
A: to identify hidden patterns, structures, and relationships within the data

Q: What are the risks of using Generative AI?
A: data privacy, intellectual property, and academic integrity
```

Certain outputs remained consistent regardless of the seed value.

Conclusion:

Changing the seed value resulted in noticeable differences when comparing outputs from distilGPT2 and GPT2. However, tasks such as summarization, question answering, and masked language modeling showed very similar results across seed values, suggesting that these tasks are less sensitive to seed variations.